

Evaluation. Teams will be evaluated on catch success rate, interception accuracy, robustness to different belt speeds, tracking quality, timing precision, and smoothness of the robot motion.

Project 3: Perception-Guided Arm Control in Unstructured Scenes

Robot: Elephant Robotics MyArm 300 Pi 7DOF robotic arm

Deployment: Real robot

In this project, teams will develop a robotic manipulation system that combines visual perception and control to interact with objects in an unstructured environment. Rather than executing predefined motions, the robot must continuously use visual feedback to adapt its behavior to changing object poses, partial occlusions, scene clutter, and sensing uncertainty.

A camera mounted in the workspace will observe a box containing multiple objects. The robot’s task is to identify and manipulate only a specific subset of target objects (for example, locating and grasping 5 rubber ducks from a box containing 20 mixed items). Not all target objects may be visible initially, requiring the system to reason about partially occluded scenes and adapt as objects move during interaction.

Possible subtasks include object detection and recognition, visual servoing, grasp planning, object alignment, and pick-and-place operations in cluttered environments. Teams must design both the perception pipeline and the robot control strategy required to reliably identify, grasp, and relocate the desired objects.

Core tasks.

- Detect and localize task-relevant objects using RGB or RGB-D sensing.
- Estimate object pose relative to the robot.
- Generate arm motions based on the perceived scene.
- Adapt online to changes or errors in perception.
- Avoid obstacles and respect robot constraints.

Relevant methods. RGB-D perception, camera calibration, pose estimation, visual servoing, inverse kinematics, optimization-based control, obstacle avoidance, and SE(3) transformations.

Evaluation. Teams will be evaluated on task success rate, perception accuracy, control accuracy, robustness to object pose variation, collision avoidance, and ability to recover from perception errors.